

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 9

(1) Research Program/Project/Study Title: MILKFISH TOCINO	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		ROXAS CITY	
Province:		CAPIZ	
Region:		VI	
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Food processing sector Restaurants Small and medium enterprise Academe		(8) Discipline/Field of Specialization Culinary	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 2 of 9

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The Philippines, a global milkfish (bangus) leader, recognizes its potential for value-added products. This study aimed to develop and evaluate the acceptability of Milkfish Tocino. By transforming milkfish into a popular meat product, it sought to promote its nutritional value and contribute to the economy.

The study determined the Acceptability of Milkfish Tocino with Natural Colorant. Specifically, it aimed to (1) determine the most preferred treatment of Milkfish Tocino (2) determine the sensory characteristics of the most preferred treatment (3) determine the consumer acceptability of the most acceptable treatment. The study was conducted at Capiz State University, Dayao Satellite College.

Results showed that Treatment C 750g Milkfish Tocino mixed with 302.46g of curing solution was evaluated as the most preferred formulation. In terms of general acceptability, it was evaluated very much. In sensory attributes, Treatment C (750g Milkfish Tocino) in terms of Color was evaluated as "Naïve Peach", in terms of Odor it was evaluated as "Desirable", in terms of Texture it was evaluated as "Extremely soft and moist", and in terms of Taste it was evaluated as "moderately sweet". The target population of the study were children and middle age adult and evaluated the sample as "Like very much". There was no significant difference among treatments of Milkfish Tocino

This study demonstrates the potential of Milkfish Tocino as a viable and acceptable product, contributing to diversifying fish-based processed foods and promoting healthier consumption. Future research can explore market potential and consumer acceptance on a larger scale.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include the process, or methodology employed.)

Milkfish frozen food, Cured milkfish

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The Philippines, alongside Indonesia and Taiwan, stands as a leading global producer of milkfish, locally known as "bangus." This aquatic staple significantly contributes to Filipino diet, health, and economy. As J. Navarro (2020) emphasized, combining proper nutrition with physical activity is crucial for achieving a healthy weight, reducing chronic disease risks like heart disease and cancer, and promoting overall well-being. Milkfish, rich in nutrients, enhances eyesight, improves cognition, protects against cancer, supports kidney and heart health, and safeguards against anxiety and depression. Its abundant vitamin D, phosphorus, and calcium are essential for strong bone development. Beyond its nutritional value, milkfish holds substantial economic importance. Past studies, such as Makmur (2020), revealed that processed fish products often generate higher profits compared to fresh ones, underscoring the economic potential of value-added fish products.

Recognizing the Philippines' position as a major milkfish producer and the economic benefits of processed fish, this study aimed to develop and evaluate the acceptability of Milkfish Tocino. By transforming a popular meat product, "tocino," into a fish-based alternative, this research sought to highlight the nutritional value of tocino while capitalizing on the country's milkfish abundance. Specifically, the study determined the most preferred Milkfish Tocino treatment, assessed its sensory characteristics (color, odor, texture, and taste), and evaluated overall consumer acceptability.

Milkfish, a globally important food species, thrives in Southeast Asia, particularly in Indonesia, Taiwan, and the Philippines. Over the past fifteen years, research efforts have intensified to address knowledge gaps in milkfish cultivation and processing. Studies like "Fisheries Science of Milkfish" (2012) and those by de Jesus-Ayson et al. (2010) and FAO (2017) highlight the fish's significance in aquaculture and its substantial production in the region.

Previous research on processed milkfish products, such as the work of C. Lim and E. Borromeo (2017) on Milkfish tapa, emphasized the importance of product innovation to meet evolving consumer preferences and market trends. Moreover, the growing recognition of the health benefits of plant-based diets and fish consumption, as supported by the

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 9

World Health Organization and studies on reduced blood pressure and cholesterol levels, further motivated the development of Milkfish Tocino.

Building on these insights, this study explored the potential of milkfish as a viable alternative to pork in tocino production. Drawing inspiration from successful initiatives in developing tilapia-based tocino and longganisa, as reported by Dr. Calanoga and Rivera (2009), this research aimed to contribute to the growing market for fish-based processed foods while promoting a healthier and more sustainable food choice.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Materials and Methods

Materials

The following are the list of materials used in this study. Measuring cups, chopping board, measuring spoon, colander, bowls, weighing scale, scissors/forcep, knife and stainless bowls

Methods

An experimental approach was used to assess the acceptability of Milkfish Tocino as a replacement for meat, chicken, or pork. To achieve this, the researcher conducted a study employing a completely randomized design. This involved creating four different versions of Milkfish Tocino, each substituting a different type of meat, and then evaluating their appearance, smell, taste, and texture.

Raw materials

Deboned Milkfish, Salt, Sugar, Monosodium Glutamate, Curing Salt, Black Pepper, Pineapple Juice, Phosphate, Bougainvillea extract and Water

Experimental Treatment

Ingredient s	Treatment A	Treatment B	Treatment C	Treatment D
Deboned Milkfish	25%	50%	75%	100%
Other Ingredient s	Ingredients are constant Other			

METHODS

Procedure in Deboning of Milkfish

The fish was washed thoroughly with clean water and drained. Its scales were removed, and the fish was washed and drained again. It was then split open from the tail to the head along the dorsal side by running a knife along the backbone. The fish was laid open like a butterfly, and its gills, internal organs, blood, and dirt were removed by thorough washing. The backbone was removed by laying the fish flat on a chopping board with the skin down. A knife was held

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 9

horizontally or slantingly and used to cut along the backbone from head to tail. The dorsal fin was trimmed, and the fish was placed flat on a tray to identify the precise location of the spines, especially the intermuscular ones.

The fish was placed in a shallow tray. With forceps, the rib bones in the belly cavity were removed. The dorsal and lateral intermuscular spines were extracted. The large arch-shaped spines at the base of the operculum were pulled out first, followed by the Y-shaped spines up to the mid-body, ending with three delicate spines. The ventral intramuscular spines were removed by making a shallow slit between the muscle segments of the ventral side, from the mid-body to the tail. The first two very fine and delicate spines found in the mid-body, which marked the beginning of the spines in this section, were pulled out.

Procedure in making Milkfish Tocino

The deboned milkfish was prepared by mixing salt and curing salt into the meat. Phosphate dissolved in water was added and mixed thoroughly. Other dry ingredients were mixed together and incorporated into the meat mixture. The liquid ingredients were then added, and everything was blended uniformly. The mixture was cured in the refrigerator for 24 hours, turning it daily to ensure even curing. Finally, it was packed in polypropylene bags and stored in the freezer until needed.

Experimental Design and Treatment

Experimental Design

The study used Complete Randomized Design (CRD) with four (4) treatments and two (2) replications. In a Completely Randomized Design (CRD) the treatment was assigned completely at random so that each experimental unit had the same chance of receiving any treatment. For the CRD, any difference among experimental units receiving the same treatment is considered as experimental error (Gomez & Gomez, 1984).

Experimental Treatment

The experiment had four treatments A (250g debone milkfish) , B (500g debone milkfish , C (750g debone milkfish) and D (1000g) Milkfish meat.

Data Collection

The main instrument used in collecting the data was sensory evaluation. It was used to evaluate the three treatments of the product. There were four (4) factors rated such as color, odor, texture, and taste. The product was evaluated by evaluators who were composed of faculty, staff, and students of the Food Technology Department of CapSu Dayao. The evaluators were oriented first before conducting the sensory evaluation. The same procedure was followed for the replication of the experiment.

Data Gathering Procedure

The data were gathered by distributing the sample product to the respondents, Evaluation sheet using score cards line scale contains nine (9) point hedonic scale for the two (2) quality attributes and were distributed to 10 trained panelists in sensory evaluation to evaluate the sensory qualities in terms of color, odor, texture, and taste. While the 9-Point Hedonic Scale was distributed to the 100 consumers to test the general acceptability of the most preferred formulation (MPF) of milkfish tocino. After the respondents answered the evaluation sheet the researchers immediately collected the data and then tabulated.

Statistical Tools and Analysis

The data were collected and collated in a tabulation sheet for analysis and interpretation. The statistical tools used in the analysis of data were mean and ANOVA. Mean was computed to determine the significant differences in the sensory qualities of Milkfish tocino among the four (4) treatments.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 9

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Results and Discussion

Results and Discussion

Table 1 Mean Rank of Most preferred formulation (MPF) Milkfish Tocino

Trial	Treatment A	Treatment B	Treatment C	Treatment D
1	2.6	3.2	1.8	2.7
2	2.5	3	2.2	2.2
Mean	2.55	3.1	2	2.45

Based on the mean rank, treatment C has the highest rank, which makes it the most preferred formulation with 750 grams of milkfish combined with other

Table 2a Friedman's Test Results of the four formulations

N	20
Chi-square	7.571
Df.	3
Asymp. Sig.	0.56

With a 5% level of confidence the result of Friedman test shows that there was no significant difference of the rank of 4 treatments. Therefore, you can use the four (4) treatments with 250 grams debone milkfish, 750 grams milkfish meat and 1000 grams debone milkfish except in Treatment B with 500 grams of debone milkfish in making the Milkfish Tocino.

Table 2b. General Acceptability of Different Treatment Milkfish Tocino with Natural Color

Treatment	Replication 1	Replication 2	Mean	Verbal Interpretation
1	7.7	7.7	7.7	LM (Like Moderately)
2	7.2	7.3	7.25	LM (Like Moderately)
3	8.2	8	8.1	LVM (Like Very Much)
4	7.9	7.6	7.75	LM (Like Moderately)

The total of (4) four different formulations are shown in table 4 In Treatment A, Formulation 1 (250 grams of debone milkfish) has 7.7, Treatment B, Formulation 2 (500 grams of debone milkfish) has a total of 7.25, Treatment C, Formulation 3 (750 grams of debone milkfish) has a total mean of 8.1, Treatment D Formulation 4 (1000 grams of debone milkfish) has a total mean of 7.75 .Therefore, treatment C got the highest acceptability level of 8.1 (750 grams of debone milkfish) with verbal interpretation of "like very much".

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 9

Table 3 Sensory characteristics of most acceptable treatment in terms of color, odor, texture and taste

	Trial 1	Trial 2	Mean	Verbal Interpreting
Color	7.05	11.32	9.18	Naïve Peach
Odor	12.06	14.82	13.71	Extremely Fishy Odor
Texture	11.58	10.30	10.94	Extremely Soft and moist
Taste	12.75	10.4	11.57	Moderately sweet

Table 3 shows that the Sensory Acceptability of Milkfish Tocino in terms of Color, Treatment C (750g debone milkfish) on the first trial had a score of 7.05 and got 11.32 on the second trial. Overall, the color of the most preferred formulation of Debone Milkfish Tocino with a mean of 9.18 which is described as “Naïve peach”.

In terms of odor, Treatment C (750g debone milkfish) on the first trial had a score of 12.06 and got 14.82 on the second trial. Overall, the odor of the most preferred formulation of Debone Milkfish Tocino with a mean of 13.71 which is described as “Extremely Fishy Odor”.

In terms of Texture, Treatment C (750g debone milkfish) on the first trial had a score of 11.58 and got 10.30 on the second trial. Overall, the texture of the most preferred formulation of debone milkfish tocino had a mean of 10. 94 which is described as “extremely soft and moist”.

In Terms of Taste, Treatment C (750g debone milkfish) on the first trial had a score of 12.75 and got 10.4 on the second trial. Overall, the taste of the most preferred formulation of debone milkfish tocino had a mean of 11.57 which is described as “Moderately Sweet”.

Therefore, the sensory characteristics of Milkfish Tocino in terms of color was “Naïve peach”, in terms of odor was “Extremely Fishy Odor”, in terms of texture was “extremely soft and moist”, and in terms of taste was “moderately sweet”.

Table 4 Consumer acceptability of the most preferred treatment

Consumers' Category	Mean	Verbal Interpretation
Child (5-12 years old)	8.76	Like Extremely
Teen (13-19 years old)	8.4	Like Extremely
Adult (20-39 years old)	8.2	Like Extremely
Middle Age Adult (40- 59 years old)	8.32	Like Extremely
Mean	8.42	Like Extremely

Results in table 7 shows the consumer test for general acceptability of the Most Preferred Formulation (MPF) Milkfish Tocino shown in Table 7. Children (5-12 years old) gained rating 8.76 like extremely, Teens (13-19 years old) gained rating 8.40 like extremely, Adult (20-39 years old) gained rating 8.20 like extremely, Middle Age Adult (40-59 years old) gained rating 8.32 like extremely. Therefore, there was no significant difference between four (4) different groups of respondents. (See Appendix B.5) Overall according to the data gathered, child, teens and adults were the study's target population since they gave the sample a "Like Extremely" rating.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 9

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSION AND RECOMMENDATIONS

The most preferred formulation (MPF) among treatments of Milkfish Tocino was Treatment C (750 grams debone milkfish). The sensory acceptability of the most preferred treatment for Milkfish Tocino was rated as "Naïve Peach" in terms of color, "Extremely fishy odor" in terms of odor, "Extremely soft and moist" in terms of texture, and "Moderately Sweet" in terms of taste.

The acceptability rating of the MPF of the Milkfish Tocino is between neither like nor dislike to like extremely among the four (4) age groups, the results shows on the data we gathered that the target consumers were child, teens and adults with a rating of "Like very much".

The use of 750g of Boneless Milkfish soaked in a 129.32g solution in making Tapa is recommended. To further evaluate the consumer acceptability of Treatment C, it is recommended to conduct a larger-scale study with a diverse group of consumers, including individuals from various age groups, cultural backgrounds, and preferences.

It is also recommended to use qualitative data, such as open-ended questions or focus group discussions, to gather more detailed feedback on the taste, texture, aroma, and appearance of Treatment C. And it would be beneficial to compare Treatment C with other existing products or variations to determine its competitive advantage and potential market positioning.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The study aimed to conduct product development on Milkfish Tocino. It was conducted at Capiz State University Fisheries and Food Research Development Center, Dayao Satellite College last May, 2024. The study used 4 treatments with three (3) replications. Ten (10) panel of experts evaluated the sensory qualities of the products, and One hundred (100) panelist for consumer test (50 college student and 50 employee).

(15) Technology Transfer Activity (Indicate the specific products, activity design, processes or services transferred as extension activities)

Production of Milkfish Tocino. Process will be taught to the target clientele in the community through extension program of the college. Present the technology to the community and intprneur. Present the study in research symposium. Milkfish Tocino for family consumption and backyard micro industry.

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

My deepest appreciation goes to the Capiz State University, FFRDC their facilities, technical support, and research environment were instrumental in the conduct of this study. Special thanks to Dr. Editha C. Alfon and Dr. Sebastian C. Caduco for funding this research

I am grateful also to Dr. Janet Ledesma for her constructive feedback and suggestions, which greatly improved the clarity and impact of our study.

Lastly, I would like to thank the participants who took part in the sensory evaluation. Their honest feedback and enthusiastic participation were essential to the success of this research.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 8 of 9

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	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 9

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